The Surjectivity of $\mathbf{exp} : \mathfrak{so}(3) \to \mathbf{SO}(3)$ and $\mathbf{exp} : M_{n\times n}(\mathbb{C}) \to \mathbf{GL}_n(\mathbb{C})$

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1 Introduction

In this short paper, we prove some properties of the matrix exponential and logarithm, guiding us to two domain-codomain pairs where the matrix exponential is surjective: $\exp : \mathfrak{so}(3) \to \mathrm{SO}(3)$ and $\exp : M_{n\times n}(\mathbb{C}) \to \mathrm{GL}_n(\mathbb{C})$.

2 Matrix Lie Groups

Let $M_{n\times m}(\Omega)$ denote the set of $n \times m$ matrices over the set Ω. Following the convention of [1], we define a matrix lie group as a topologically closed subgroup of $\mathrm{GL}_n(\mathbb{C})$. All norms induce the same open sets in finite-dimensional real vector spaces (see §1.4 of [2]), so we choose the operator norm: $\|\mathbf{X}\|_{\text{op}} := \sup\{\|\mathbf{Xv}\|_2 : \|\mathbf{v}\|_2 = 1\}$, where $\|\cdot\|_2$ is the Euclidean norm on $\mathbb{C}^n$.

3 The Matrix Exponential

Given a matrix $\mathbf{A} \in M_{n\times n}(\mathbb{C})$, we define the matrix exponential as

$$\exp(\mathbf{A}) := e^{\mathbf{A}} := \sum_{k=0}^{\infty} \frac{\mathbf{X}^k}{k!}.$$

This extends the familiar definition of exponentiation from $\mathbb{C}$ to square matrices over $\mathbb{C}$. Before we can prove that this series converges, we need two facts. The first is Theorem 1.3.9 of *An Introduction to Banach Space Theory* [3]. We restate the theorem below but skip the proof for brevity.

Proposition 1. *Consider any series* $\mathbf{s}_m = \sum_{k=0}^{m} \mathbf{x}_k$ *in a complete normed vector space* $(X, \|\cdot\|)$*. If* $\sum_{k=0}^{\infty} \|\mathbf{x}_k\| < \infty$ *(i.e. the series converges absolutely), then* $\mathbf{s}_m$ *converges in* $(X, \|\cdot\|)$*.*

This allows us to prove the convergence of $e^{\mathbf{A}}$ by showing that it is absolutely convergent. Absolute convergence follows from our next proposition.

Proposition 2. *If* $\mathbf{X}, \mathbf{Y}$ *are any two elements of* $M_{n\times n}(\mathbb{C})$*, then* $\|\mathbf{XY}\|_{op} \le \|\mathbf{X}\|_{op}\|\mathbf{Y}\|_{op}$*.*

Proof: Recall that $\|\mathbf{XY}\|_{\text{op}} = \sup\{\|\mathbf{XYv}\|_2 : \|\mathbf{v}\|_2 = 1\}$. Let $\mathbf{u} = \mathbf{Yv}$. Then $\|\mathbf{u}\|_2 \le \|\mathbf{Y}\|_{\text{op}}\|\mathbf{v}\|_2 = \|\mathbf{Y}\|_{\text{op}}$ by the definition of the operator norm. Likewise, $\|\mathbf{Xu}\|_2 \le \|\mathbf{X}\|_{\text{op}}\|\mathbf{u}\|_2 \le \|\mathbf{X}\|_{\text{op}}\|\mathbf{Y}\|_{\text{op}}$. Combining the previous two inequalities yields $\|\mathbf{XYv}\|_2 \le \|\mathbf{X}\|_{\text{op}}\|\mathbf{Y}\|_{\text{op}}$ for all $\mathbf{v} \in \mathbb{C}^n$ with unit norm. Thus, the supremum over all such $\mathbf{v}$ will be less than or equal to $\|\mathbf{X}\|_{\text{op}}\|\mathbf{Y}\|_{\text{op}}$. $\square$

We can now easily show that $e^{\mathbf{A}}$ is convergent for any square matrix over $\mathbb{C}$. Note that $\|\mathbf{A}^m\|_{\text{op}} \le \|\mathbf{A}\|_{\text{op}}^m$ by Proposition 2, so

$$\sum_{k=0}^{\infty} \left\|\frac{\mathbf{A}^k}{k!}\right\|_{\text{op}} \le \sum_{k=0}^{\infty} \frac{1}{k!}\|\mathbf{A}\|_{\text{op}}^k = e^{\|\mathbf{A}\|_{\text{op}}} < \infty.$$

So $e^{\mathbf{A}}$ converges by Proposition 1. Furthermore, this function is continuous since for any closed ball of radius $r > 0$ centered around $\mathbf{0}$ in $M_{n\times n}(\mathbb{C})$,

$$\left\|e^{\mathbf{X}} - \sum_{k=0}^{n} \frac{\mathbf{X}^k}{k!}\right\|_{\text{op}} = \left\|\sum_{k=0}^{\infty} \frac{\mathbf{X}^k}{k!} - \sum_{k=0}^{n} \frac{\mathbf{X}^k}{k!}\right\|_{\text{op}} = \left\|\sum_{k=n+1}^{\infty} \frac{\mathbf{X}^k}{k!}\right\|_{\text{op}} \le \sum_{k=n+1}^{\infty} \frac{1}{k!}\|\mathbf{X}\|_{\text{op}}^k \le \sum_{k=n+1}^{\infty} \frac{r^k}{k!}.$$

Since $\sum_{k=0}^{\infty} \frac{r^k}{k!}$ converges to e^r, the tail end of the partial sums must approach zero, meaning that for any $\varepsilon > 0$, we can choose an n_ε large enough that $\sum_{k=n_\varepsilon+1}^{\infty} \left(\frac{r^k}{k!}\right) < \varepsilon$. Since this is true for all $\mathbf{X}$ with $\|\mathbf{X}\|_{\text{op}} \leq r$, the partial sums converge uniformly to $e^{\mathbf{X}}$ on this set. Since any compact set in $M_{n\times n}(\mathbb{C})$ can be covered by a closed ball of large enough radius, this means the series converges uniformly on all compact subsets of $M_{n\times n}(\mathbb{C})$. That means that the series is normally convergent, so the fact that the partial sums are continuous means that the limit function is continuous (See §5.2 of [4]).

Proposition 3. *For any $\mathbf{X}, \mathbf{Y} \in M_{n\times n}(\mathbb{C})$, the following facts hold.*

(a) *if $\mathbf{X}$ and $\mathbf{Y}$ commute, then $e^{\mathbf{X}+\mathbf{Y}} = e^{\mathbf{X}}e^{\mathbf{Y}}$.*

(b) *if $\mathbf{C}$ is invertible, then $e^{\mathbf{C}\mathbf{X}\mathbf{C}^{-1}} = \mathbf{C}e^{\mathbf{X}}\mathbf{C}^{-1}$.*

Proof of part (a): Because the matrix exponential converges absolutely, we can multiply the series term by term according to Merten's Theorem (Theorem 3.50 of [5]), so

$$e^{\mathbf{X}}e^{\mathbf{Y}} = (\mathbf{I} + \mathbf{X} + \frac{\mathbf{X}^2}{2!} + \frac{\mathbf{X}^3}{3!} + \cdots)(\mathbf{I} + \mathbf{Y} + \frac{\mathbf{Y}^2}{2!} + \frac{\mathbf{Y}^3}{3!} + \cdots).$$

The terms of combined degree m are the product of an $\mathbf{X}$ term of degree $k \leq m$ and a $\mathbf{Y}$ term of degree $m-k$, so the product simplifies to

$$e^{\mathbf{X}}e^{\mathbf{Y}} = \sum_{m=0}^{\infty}\sum_{k=0}^{m} \frac{\mathbf{X}^k\mathbf{Y}^{m-k}}{k!(m-k)!} = \sum_{m=0}^{\infty} \frac{1}{m!}\sum_{k=0}^{m} \frac{m!\mathbf{X}^k\mathbf{Y}^{m-k}}{k!(m-k!)} = \sum_{m=0}^{\infty} \frac{1}{m!}\sum_{k=0}^{m} \binom{m}{k}\mathbf{X}^k\mathbf{Y}^{m-k}. \tag{3.1}$$

In (3.1) we have what looks like the binomial theorem. The binomial theorem assume that we can choose products in any order, i.e. that $\mathbf{X}$ and $\mathbf{Y}$ commute. Luckily, they do, so

$$e^{\mathbf{X}}e^{\mathbf{Y}} = \sum_{m=0}^{\infty} \frac{1}{m!}(\mathbf{X}+\mathbf{Y})^m = e^{\mathbf{X}+\mathbf{Y}}.$$

Proof of part (b): The key is to recognize that $(\mathbf{C}\mathbf{X}\mathbf{C}^{-1})^n = \mathbf{C}\mathbf{X}^n\mathbf{C}^{-1}$. Thus,

$$\begin{aligned} e^{\mathbf{C}\mathbf{X}\mathbf{C}^{-1}} &= \lim_{m\to\infty}\sum_{n=0}^{m} \frac{1}{n!}(\mathbf{C}\mathbf{X}\mathbf{C}^{-1})^n \\ &= \lim_{m\to\infty}\sum_{n=0}^{m} \frac{1}{n!}\mathbf{C}\mathbf{X}^n\mathbf{C}^{-1} \\ &= \mathbf{C}\Big(\lim_{m\to\infty}\sum_{n=0}^{m} \frac{1}{n!}\mathbf{X}^n\Big)\mathbf{C}^{-1} \\ &= \mathbf{C}e^{\mathbf{X}}\mathbf{C}^{-1}. \quad \square \end{aligned} \tag{3.2}$$

Proposition 3 is a powerful tool for computing exponentials and will aid us in proving the following results.

4 The Matrix Logarithm

Just like for the exponential, we can use our existing definitions from the complex numbers to extend the logarithm to $M_{n\times n}(\mathbb{C})$. For any $\mathbf{A} \in M_{n\times n}(\mathbb{C})$ such that $\|\mathbf{A}-\mathbf{I}\|_{\text{op}} < 1$, we define the matrix logarithm as

$$\log(\mathbf{A}) = \sum_{m=1}^{\infty} (-1)^{m+1}\frac{(\mathbf{A}-\mathbf{I})^m}{m}.$$

If we consider the absolute convergence of $\log \mathbf{A}$, we can see that

$$\sum_{m=0}^{\infty} \left\|(-1)^{m+1}\frac{(\mathbf{A}-\mathbf{I})^m}{m}\right\|_{\text{op}} = \sum_{m=0}^{\infty} \frac{1}{m}\left\|(\mathbf{A}-\mathbf{I})^m\right\|_{\text{op}} \leq \sum_{m=0}^{\infty} \|\mathbf{A}-\mathbf{I}\|_{\text{op}}^m. \tag{4.1}$$

Since $\|\mathbf{A}-\mathbf{I}\|_{\text{op}} < 1$, (4.1) converges by the geometric series test. So $\log \mathbf{A}$ converges by Proposition 1.

Proposition 4. *log is continuous on the set* $\{\mathbf{A} \in M_{n\times n}(\mathbb{C}) : \|\mathbf{A}-\mathbf{I}\|_{op} < 1\}$.

Proof: Let $\mathbf{X} = \mathbf{A} - \mathbf{I}$. Then by assumption, $\|\mathbf{X}\|_{\mathrm{op}} < 1$, and

$$\log \mathbf{A} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}(\mathbf{A}-\mathbf{I})^n = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}\mathbf{X}^n.$$

Let $R \in (0,1)$. Let $\|\mathbf{X}\|_{\mathrm{op}} \leq R$. Then

$$\begin{aligned}\Big\| \log \mathbf{A} - \sum_{n=1}^{m} \frac{(-1)^{n+1}}{n}\mathbf{X}^n \Big\|_{\mathrm{op}} &= \Big\| \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}\mathbf{X}^n - \sum_{n=1}^{m} \frac{(-1)^{n+1}}{n}\mathbf{X}^n \Big\|_{\mathrm{op}} \\ &= \Big\| \sum_{n=m+1}^{\infty} \frac{(-1)^{n+1}}{n}\mathbf{X}^n \Big\|_{\mathrm{op}} \\ &\leq \sum_{n=m+1}^{\infty} \Big| \frac{(-1)^{n+1}}{n} \Big| \|\mathbf{X}\|_{\mathrm{op}}^n \\ &\leq \sum_{n=m+1}^{\infty} \Big| \frac{(-1)^{n+1}}{n} \Big| R^n.\end{aligned} \tag{4.2}$$

Since $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} R^n = \log(R+1)$, we know that the tail of the sum approaches 0, so as $m \to \infty$, the last line of (4.1) gets arbitrarily small. That is, for any $\varepsilon > 0$, there is some positive integer M such that $\sum_{n=M+1}^{\infty} \frac{(-1)^n+1}{n} R^n < \varepsilon$. Since this bound does not depend on which $\mathbf{X}$ we choose, the series converges uniformly to $\log \mathbf{A}$ on the comapct set $\{\|\mathbf{A}-\mathbf{I}\|_{\mathrm{op}} \leq R\}$. This is true for all $R \in (0,1)$, so the series converges normally on $\{\|\mathbf{A}-\mathbf{I}\|_{\mathrm{op}} < 1\}$. Since each of the partial sums is a continuous polynomial and the convergence is normal, $\log \mathbf{A}$ is continuous on $\{\|\mathbf{A}-\mathbf{I}\|_{\mathrm{op}} < 1\}$. $\square$

Proposition 5. *For all* $\mathbf{A}, \mathbf{B} \in M_{n\times n}(\mathbb{C})$*:*

(a) *If* $\|\mathbf{A}-\mathbf{I}\|_{op} < 1$*, then* $\exp(\log \mathbf{A}) = \mathbf{A}$.

(b) *If* $\|\mathbf{B}\|_{\mathrm{op}} < \log 2$, then $\log(\exp \mathbf{B}) = \mathbf{B}$.

Proof of part (a): Let $\mathbf{X} = \mathbf{A} - \mathbf{I}$. Then $\exp(\log \mathbf{A}) = \mathbf{A}$ if and only if $\exp(\log(\mathbf{I}+\mathbf{X})) = \mathbf{I}+\mathbf{X}$. We break this proof down into two cases.

Case 1: X is diagonalizable. Then $\mathbf{X} = \mathbf{C}\mathbf{D}\mathbf{C}^{-1}$ where $\mathbf{D}$ is a diagonal matrix whose entries are the (not necessarily distinct) eigenvalues of $\mathbf{A}$. Since $\|\mathbf{X}\|_{\mathrm{op}} < 1$ by assumption, each eigenvalue λ_i has modulus less than 1 (since otherwise $\mathbf{X}$ scales some vector by more than $\|\mathbf{X}\|_{\mathrm{op}}$, which is impossible). Thus, $\lambda_i + 1$ is contained in the ball of radius 1 around 1 in $\mathbb{C}$, so $\log(\lambda_i + 1)$ converges. So

$$\begin{aligned}\log(\mathbf{X}+\mathbf{I}) &= \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m}\mathbf{X}^m \\ &= \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m}(\mathbf{C}\mathbf{D}\mathbf{C}^{-1})^m \\ &= \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m}\mathbf{C}\begin{pmatrix}\lambda_1^m & 0 & \dots & 0 \\ 0 & \lambda_2^m & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n^m\end{pmatrix}\mathbf{C}^{-1} \\ &= \mathbf{C}\begin{pmatrix}\log(\lambda_1+1) & 0 & \dots & 0 \\ 0 & \log(\lambda_2+1) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \log(\lambda_n+1)\end{pmatrix}\mathbf{C}^{-1}.\end{aligned} \tag{4.3}$$

Since $|(\lambda_i+1)-1|<1$, $e^{\log(\lambda_i+1)} = \lambda_i+1$. Thus,

$$\begin{aligned}
\exp(\log(\mathbf{X}+\mathbf{I})) &= \mathbf{C}\begin{pmatrix} \lambda_1+1 & 0 & \dots & 0 \\ 0 & \lambda_2+1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n+1 \end{pmatrix}\mathbf{C}^{-1} \\
&= \mathbf{C}(\mathbf{D}+\mathbf{I})\mathbf{C}^{-1} \\
&= \mathbf{C}\mathbf{D}\mathbf{C}^{-1}+\mathbf{I} \\
&= \mathbf{X}+\mathbf{I}.
\end{aligned} \tag{4.4}$$

Case 2: X is not diagonalizable. Then since the characteristic polynomial of $\mathbf{X}$ splits over $\mathbb{C}$, it has a Jordan decomposition $\mathbf{X} = \mathbf{B}\mathbf{J}\mathbf{B}^{-1}$, where $\mathbf{J}$ has the eigenvalues of $\mathbf{X}$ on its diagonal. Let $\mathbf{D}$ be the diagonal matrix with diagonal entries $1, 2, \dots, n$. Then let $\mathbf{X}_k = \mathbf{B}(\mathbf{J}+\frac{1}{k}\mathbf{D})\mathbf{B}^{-1}$ define a sequence that converges to $\mathbf{X}$. $\mathbf{X}_k$ is written in Jordan normal form (since the addition of $\frac{1}{k}\mathbf{D}$ only changes the diagonal entries). Thus, the i^{th} eigenvalue of $\mathbf{X}_k$ is $\mathbf{J}_{ii}+\frac{i}{k}$. For sufficiently large k, each of these values will be distinct, meaning $\mathbf{X}_k$ will be diagonalizable. Furthermore, since $\mathbf{X}_k \to \mathbf{X}$ and $\|\mathbf{X}\|_{\text{op}} < 1$, eventually $\|\mathbf{X}_k\|_{\text{op}} < 1$. By the continuity of log when $\|\mathbf{X}\|_{\text{op}} < 1$ and the global continuity of exp, we can swap limits for this sequence for large k, so

$$\exp(\log(\mathbf{X}+\mathbf{I})) = \exp(\log(\lim_{k\to\infty}[\mathbf{X}_k+\mathbf{I}])) = \lim_{k\to\infty}\exp(\log(\mathbf{X}_k+\mathbf{I})) = \lim_{k\to\infty}\mathbf{X}_k+\mathbf{I} = \mathbf{X}+\mathbf{I}. \tag{4.5}$$

Where we were able to cancel exp and log in the fourth expression using the diagonalizable case.

Proof of part (b): First we note that if $\|\mathbf{B}\|_{\text{op}} < \log 2$, then

$$\begin{aligned}
\|e^{\mathbf{B}}-\mathbf{I}\|_{\text{op}} &= \|(\mathbf{I}+\mathbf{B}+\frac{\mathbf{B}^2}{2!}+\frac{\mathbf{B}^3}{3!}+\cdots)-\mathbf{I}\|_{\text{op}} \\
&\le \|\mathbf{B}\|_{\text{op}} + \left\|\frac{\mathbf{B}^2}{2!}\right\|_{\text{op}} + \left\|\frac{\mathbf{B}^3}{3!}\right\|_{\text{op}} + \cdots \\
&= e^{\|\mathbf{B}\|_{\text{op}}}-1 \\
&< e^{\log 2}-1 \\
&= 1.
\end{aligned} \tag{4.6}$$

Again, if $\mathbf{B}$ is diagonalizable, then $\mathbf{B} = \mathbf{C}\mathbf{D}\mathbf{C}^{-1}$, where $\mathbf{D}$ is a diagonal matrix of the eigenvalues $\lambda_1, \dots, \lambda_n$ of $\mathbf{B}$ and $\mathbf{C}$ is invertible. Since the exponential of a diagonal matrix is taken entry-wise, $e^{\mathbf{D}}_{ii} = e^{\lambda_i}$. Furthermore, these are the eigenvalues of $e^{\mathbf{B}}$ because $e^{\mathbf{B}} = \mathbf{C}e^{\mathbf{D}}\mathbf{C}^{-1}$ is a diagonalization of $e^{\mathbf{B}}$.To get a bound on the modulus of these eigenvalues, consider that subtracting $\mathbf{I}$ from a matrix decreases all its eigenvalues by 1, so the eigenvalues of $e^{\mathbf{B}}-\mathbf{I}$ are $e^{\lambda_i}-1$. Since $\|e^{\mathbf{B}}-\mathbf{I}\|_{\text{op}} < 1$, each eigenvalue must have modulus less than 1 as shown in case 1 of part (a). Thus, $|e^{\lambda_i}-1| < 1$, so $\log e^{\lambda_i} = \lambda_i$. Since the log of a diagonal matrix is also taken entry-wise [as we showed in equation (4.2)], that means $\log(e^{\mathbf{D}}) = \mathbf{D}$. In (4.2), we also showed that $\log(\mathbf{C}\mathbf{D}\mathbf{C}^{-1}) = \mathbf{C}\log(\mathbf{A})\mathbf{C}^{-1}$ for invertible $\mathbf{C}$ and diagonal $\mathbf{A}$. Thus,

$$\begin{aligned}
\log e^{\mathbf{B}} &= \log(\mathbf{C}e^{\mathbf{D}}\mathbf{C}^{-1}) \\
&= \mathbf{C}\log(e^{\mathbf{D}})\mathbf{C}^{-1} \\
&= \mathbf{C}\mathbf{D}\mathbf{C}^{-1} \\
&= \mathbf{B}.
\end{aligned} \tag{4.7}$$

In the case that $\mathbf{B}$ is not diagonalizable, we can approximate it by a sequence of diagonalizable matrices like in part (a), using the continuity of log and exp to get the desired result. □

5 Cases of Surjectivity of The Exponential

We now have the tools needed to investigate cases where the exponential map is surjective.

5.1 Surjectivity of $\exp : M_{n\times n}(\mathbb{C}) \to \mathbf{GL}_n(\mathbb{C})$

To prove that this map is surjective, we follow the outline of exercises 2.8 and 2.9 from *Lie Groups, Lie Algebras, and Representations* [1]. First we require some definitions. A square matrix $\mathbf{N}$ is said to be nilpotent if there exists a positive integer k such that $\mathbf{N}^k = \mathbf{0}$. A square matrix $\mathbf{U}$ is unipotent if $\mathbf{U} - \mathbf{I}$ is nilpotent.

Proposition 6. *If* $\mathbf{U}$ *is unipotent, then* $\log(\mathbf{U})$ *is nilpotent, and if* $\mathbf{N}$ *is nilpotent, then* $\exp(\mathbf{N})$ *is unipotent.*

Proof: We first must show that finite sums of commuting nilpotent matrices are nilpotent. Let $\mathbf{N}, \mathbf{M}$ be nilpotent matrices that commute with eachother. Then there exists some $k \in \mathbb{N}$ such that $\mathbf{N}^k = \mathbf{M}^k = \mathbf{0}$. By the binomial theorem: $(\mathbf{N}+\mathbf{M})^p = \sum_{j=1}^{p} \binom{p}{j} \mathbf{N}^j \mathbf{M}^{p-j}$ (this is where we needed commuting matrices). If we choose $p \geq 2k$, then $p-j \geq k$ or $j \geq k$ since $(p-j)+j = p \geq 2k$. Thus, for every term in $(\mathbf{N}+\mathbf{M})^p$, either $\mathbf{N}^j$ or $\mathbf{M}^{p-j}$ is $\mathbf{0}$, meaning the whole sum is $\mathbf{0}$. So, $\mathbf{N}+\mathbf{M}$ is nilpotent. Continuing inductively, we can see this is true for any finite sum (since any term in the sum will commute with the other terms grouped together).

Now we can prove the proposition. Since $\mathbf{U}$ is unipotent, $\mathbf{U} - \mathbf{I}$ is nilpontent, so there are finitely many $k \in \mathbb{N}$ such that $(\mathbf{U} - \mathbf{I})^k \neq \mathbf{0}$. The power series of $\log(\mathbf{U})$ becomes

$$\log(\mathbf{U}) = \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} (\mathbf{U} - \mathbf{I})^m = \sum_{m=1}^{k} \frac{(-1)^{m+1}}{m} (\mathbf{U} - \mathbf{I})^m. \tag{5.1}$$

Where for the last equality we used the fact that eventually every summand is $\mathbf{0}$. Since each term in the sum is a scaled positive power of $\mathbf{U} - \mathbf{I}$, the terms all commute. Thus, this sum is nilpotent, proving part one of the proposition.

For part two, since $\mathbf{N}$ is nilpotent, there are finitely many $k \in \mathbb{N}$ such that $\mathbf{N}^k \neq \mathbf{0}$. The power series expansion of $\exp(\mathbf{N})$ can be rewritten as

$$\exp(\mathbf{N}) = \sum_{m=0}^{\infty} \frac{1}{m!} \mathbf{N}^m = \sum_{m=0}^{k} \frac{1}{m!} \mathbf{N}^m = \mathbf{I} + \sum_{m=1}^{k} \frac{1}{m!} \mathbf{N}^m. \tag{5.2}$$

Since the terms of the summation are all scaled positive powers of $\mathbf{N}$, they commute, so the summation is nilpotent. Thus, $\exp(\mathbf{N})$ is unipotent. $\square$

These facts help us expand the cases where log is the inverse of exp, as the next proposition will show.

Proposition 7. *If* $\mathbf{U}$ *is unipotent, then* $\exp(\log \mathbf{U}) = \mathbf{U}$*, and if* $\mathbf{N}$ *is nilpotent, then* $\log(\exp \mathbf{N}) = \mathbf{N}$*.*

Proof:. Let $\gamma(t) = \mathbf{I} + t(\mathbf{U} - \mathbf{I})$ be a smooth curve in $M_{n\times n}(\mathbb{C})$ parameterized by $t \in \mathbb{R}$. By Proposition 6, since $\mathbf{U}$ is unipotent, the power series of $\log \mathbf{U}$ is finite, hence a polynomial of t. Since $\log \mathbf{U}$ is nilpotent, the power series of $\exp \log(\mathbf{U})$ is finite by Proposition 6, so it is a polynomial over t. Furthermore, $\|\gamma(t) - \mathbf{I}\|_{\text{op}} = \|t(\mathbf{U} - \mathbf{I})\|_{\text{op}}$, which approaches zero as t does. Thus, we can pick some $\varepsilon > 0$ such that for all $t \in [0, \varepsilon] : \|\gamma(t) - \mathbf{I}\|_{\text{op}} < 1$. Thus, $\exp(\log \gamma(t)) = \gamma(t)$ on $[0, \varepsilon]$ by Proposition 5. If we consider $[0, \varepsilon]$ to be a subset of $\mathbb{C}$, then for the i^{th} row and j^{th} column of our matrices, $\exp \log(\gamma(t))_{i,j}$ and $\gamma(t)_{i,j}$ are two (analytic) polynomials over $\mathbb{C}$ that agree on $[0, \varepsilon]$, which has a nonisolated point. By the identity principle (See §5.7 of [4]), this means $\exp \log(\mathbf{U}(t))_{i,j} = \mathbf{U}(t)_{i,j}$ for all $t \in \mathbb{C}$. Since all entries agree, $\exp(\log(\gamma(t)) = \gamma(t)$ on all of $\mathbb{C}$, namely at $t = 1$. Since $\gamma(1) = \mathbf{U}$, we have $\exp(\log \mathbf{U}) = \mathbf{U}$ for all unipotent matrices.

The proof proceeds similarly for $\mathbf{N}$. If we let $\alpha(t) = t\mathbf{N}$, then as a consequence of Proposition 7, $\log(\exp \alpha(t))$ depends polynomially on t. Since $\|\mathbf{N}(t)\|_{\text{op}} = |t| \|\mathbf{N}\|_{\text{op}}$, we can again choose a small $\varepsilon > 0$ such that for all $t \in [0, \varepsilon] : \|\alpha(t)\|_{\text{op}} < \log(2)$. Thus, on $[0, \varepsilon]$, $\log(\exp \alpha(t)) = \alpha(t)$. Applying the identity principle element-wise as done above will yield the desired result. $\square$

With Propositions 5 and 6 under our belts, we are ready to prove $\exp : M_{n\times n}(\mathbb{C}) \to \text{GL}_n(\mathbb{C})$ is surjective. From the fundamental theorem of algebra, the characteristic polynomial of any matrix $\mathbf{A} \in \text{GL}_n(\mathbb{C})$ will split over $\mathbb{C}$. Thus, there is a Jordan decomposition of the form $\mathbf{A} = \mathbf{CJC}^{-1}$ where $\mathbf{C}$ is an invertible matrix and $\mathbf{J}$ is a Jordan block matrix (see §7 of [6]). Jordan block matrices are of the form

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}_1 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{J}_2 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{J}_k \end{bmatrix}. \tag{5.3}$$

Where each $\mathbf{J}_i = \lambda_i \mathbf{I} + \mathbf{N}_i$ for a nilpotent matrix $\mathbf{N}_i$ and a nonzero complex number λ_i (the λ_i are nonzero since $\mathbf{A}$ is invertible). We first show that any Jordan block is in the image of the exponential, then show that combining the Jordan block preimages results in a preimage for $\mathbf{J}$. Let $\mathbf{B} = \lambda\mathbf{I} + \mathbf{N} = \lambda\mathbf{I}(\mathbf{I} + \mathbf{N}/\lambda)$ be any Jordan block. Since $\lambda \neq 0$, there exists some $z \in \mathbb{C}$ such that $\lambda = e^z$. Thus, $\lambda\mathbf{I} = \exp(z\mathbf{I})$ (This is because exponentials of diagonal matrices are taken entry-wise). So we have $\mathbf{B} = \exp(z\mathbf{I})(\mathbf{I} + \mathbf{N}/\lambda)$. Since $(\mathbf{I} + \mathbf{N}/\lambda)$ is unipotent, $\exp(\log(\mathbf{I} + \mathbf{N}/\lambda)) = \mathbf{I} + \mathbf{N}/\lambda$ by Proposition 7. Thus, $\mathbf{B} = \exp(z\mathbf{I})\exp(\log(\mathbf{I} + \mathbf{N}/\lambda))$. Scalar multiples of the identity matrix commute with any other matrix , so $\mathbf{B} = \exp(z\mathbf{I} + \log(\mathbf{I} + \mathbf{N}/\lambda))$ by Proposition 3. Thus, any invertible Jordan block is in the image of the exponential on $M_{n\times n}(\mathbb{C})$.

If we let $\mathbf{L}_i = z_i\mathbf{I} + \log(\mathbf{I} + \mathbf{N}_i/\lambda_i)$ be the preimage of $\mathbf{J}_i$, then substituting $\mathbf{L}_i$ into the position of $\mathbf{J}_i$ for each $i \in \{1, \dots, k\}$ yields a matrix $\mathbf{L}$ whose image under exp is $\mathbf{J}$. This is because every power of $\mathbf{L}$ looks like

$$\mathbf{L}^m = \begin{bmatrix} \mathbf{L}_1^m & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{L}_2^m & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{L}_k^m \end{bmatrix}. \tag{5.4}$$

Taking the limit of the partial sums of $\exp(\mathbf{L})$ yields

$$\exp(\mathbf{L}) = \lim_{m\to\infty} \begin{bmatrix} \mathbf{L}_1^m/m! & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{L}_2^m/m! & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{L}_k^m/m! \end{bmatrix} = \begin{bmatrix} \exp(\mathbf{L}_1) & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \exp(\mathbf{L}_2) & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \exp(\mathbf{L}_k) \end{bmatrix}. \tag{5.5}$$

Where we used the fact that limits of matrices are taken element wise. Since $\exp(\mathbf{L_i}) = \mathbf{J}_i$, this matrix is equal to $\mathbf{J}$. Finally, since $\mathbf{C}$ is invertible, $\exp(\mathbf{CLC}^{-1}) = \mathbf{C}\exp(\mathbf{L})\mathbf{C}^{-1} = \mathbf{CJC}^{-1} = \mathbf{A}$ by Proposition 3, concluding the proof that $\exp : M_{n\times n}(\mathbb{C}) \to \mathrm{GL}_n(\mathbb{C})$ is surjective $\square$

5.2 Surjectivity of $\exp : \mathfrak{so}(3) \to \mathbf{SO}(3)$

The following definition is taken from §2.5 of *Lie Groups, Lie Algebras, and Representations* [1]. Given a matrix Lie Group G, its Lie algebra, denoted $\mathfrak{g}$, is the set of matrices $\mathbf{X}$ such that for all $t \in \mathbb{R}$: $e^{t\mathbf{X}} \in G$. The Lie algebra is of special interest in group theory because it is a vector space under the bracket operation, enabling us to study Lie groups via their Lie algebras using the tools of linear algebra [1]. While we won't explore Lie algebras deeply, we *are* concerned with the image of the exponential on the lie algebra of SO(3), which we denote $\mathfrak{so}(3)$. As shown in section 5.4 of [7], the lie algebra of $\mathfrak{so}(3)$ is the set of skew-symmetric real matrices.

Proposition 8. *The map* $\exp : \mathfrak{so}(3) \to SO(3)$ *is surjective.*

Proof: In the standard basis, a rotation through the x-axis by angle θ can be represented in the form

$$\mathbf{R}_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix}. \tag{5.6}$$

By Euler's Rotation Theorem (Euler 1775, [8]), any matrix $\mathbf{R}$ in $SO(3)$ is equivalent to a rotation about an axis spanned by a unit vector $\mathbf{v} \in \mathbb{R}^3$. If we consider any set of linearly independent vectors where $\mathbf{v}$ is our first vector, then the Graham Schmidt process will generate an orthonormal basis $\{\mathbf{v}, \mathbf{u}, \mathbf{w}\}$.

In this basis, $\mathbf{v}$ remains fixed under transformation by $\mathbf{R}$, and $\mathbf{u}, \mathbf{v}$ span the plane which $\mathbf{R}$ induces a rotation of angle θ through. Thus, in our new basis, we may represent $\mathbf{R}$ in the form of (5.6). In the language of similarity, $\mathbf{R} = \mathbf{B}\mathbf{R}_x\mathbf{B}^{-1}$ for an invertible change-of-basis matrix $\mathbf{B}$. We would now like to show that $\mathbf{R}_x$ is in the image of exp. Consider the matrix

$$\mathbf{X} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -\theta \\ 0 & \theta & 0 \end{pmatrix}.$$

The powers of $\mathbf{X}$ are cyclic, since

$$\begin{aligned}
\mathbf{X}^2 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\theta^2 & 0 \\ 0 & 0 & -\theta^2 \end{pmatrix} \\
\mathbf{X}^3 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & \theta^3 \\ 0 & -\theta^3 & 0 \end{pmatrix} \\
\mathbf{X}^4 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & \theta^4 & 0 \\ 0 & 0 & \theta^4 \end{pmatrix} \\
\mathbf{X}^5 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -\theta^5 \\ 0 & \theta^5 & 0 \end{pmatrix} = \theta^4\mathbf{X}.
\end{aligned}$$

Using this fact we can show that the explicit formula for $e^{\mathbf{X}}$ is

$$e^{\mathbf{X}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \gamma & \delta \end{pmatrix}. \tag{5.7}$$

Where the 1 in the first entry comes from the identity term $\mathbf{X}^0 = \mathbf{I}$, and the remaining terms come from inspecting the entries of successive powers of $\mathbf{X}$. Computing them yields

$$\begin{aligned}
\alpha &= 1 + \frac{0}{1!} - \frac{\theta^2}{2!} + \frac{0}{3!} + \frac{\theta^4}{4!} + \cdots = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots = \cos(\theta), \\
\beta &= 0 - \frac{\theta}{1!} + \frac{0}{2!} + \frac{\theta^3}{3!} + \frac{0}{4!} + \cdots = -\theta + \frac{\theta^3}{3!} - \frac{\theta^5}{5!} + \frac{\theta^7}{7!} + \cdots = -\sin(\theta), \\
\gamma &= 0 + \frac{\theta}{1!} + \frac{0}{2!} - \frac{\theta^3}{3!} + \frac{0}{4!} + \cdots = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots = \sin(\theta), \\
\delta &= 1 + \frac{0}{1!} - \frac{\theta^2}{2!} + \frac{0}{3!} + \frac{\theta^4}{4!} + \cdots = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots = \cos(\theta).
\end{aligned} \tag{5.8}$$

Which shows that $e^{\mathbf{X}} = \mathbf{R}_x$. Thus,

$$e^{(\mathbf{B}\mathbf{X}\mathbf{B}^{-1})} = \mathbf{B}e^{\mathbf{X}}\mathbf{B}^{-1} = \mathbf{B}\mathbf{R}_x\mathbf{B}^{-1} = \mathbf{R}. \tag{5.9}$$

If we can show that $\mathbf{B}\mathbf{X}\mathbf{B}^{-1}$ is skew-symmetric, then we are done. Recall that $\mathbf{B}$ is the change of basis matrix from the standard basis to the orthonormal basis $\{\mathbf{v}, \mathbf{u}, \mathbf{w}\}$. These are two orthonormal bases, so $\mathbf{B}$ is orthogonal (see §14.3 of [9]). So $\mathbf{B}^{\intercal} = \mathbf{B}^{-1}$ and

$$\begin{aligned}
\mathbf{B}\mathbf{X}\mathbf{B}^{-1} &= \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -\theta \\ 0 & \theta & 0 \end{pmatrix} \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix} \\
&= \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ -\theta c & -\theta f & -\theta i \\ \theta b & \theta e & \theta h \end{pmatrix} \\
&= \begin{pmatrix} \theta bc + \theta bc & -\theta bf + \theta ec & -\theta bi + \theta ch \\ -\theta ec + \theta bf & -\theta ef + \theta ef & -\theta ei + \theta fh \\ -\theta ch + \theta bi & -\theta fh + \theta ei & -\theta hi + \theta hi \end{pmatrix} \\
&= \begin{pmatrix} 0 & -\theta bf + \theta ec & -\theta bi + \theta ch \\ -(-\theta bf + \theta ec) & 0 & -\theta ei + \theta fh \\ -(-\theta bc + \theta ch) & -(\theta ei + \theta fh)ei & 0 \end{pmatrix}.
\end{aligned} \tag{5.10}$$

So $\mathbf{BXB}^{-1}$ is skew-symmetric, concluding the proof that $\exp : \mathfrak{so}(3) \to \mathrm{SO}(3)$ is surjective. $\square$

References

[1] Brian C. Hall, *Lie Groups, Lie Algebras, and Representations*, Springer-Verlag New York Inc, 2010, https://doi.org/10.1007/978-0-387-21554-9.

[2] Amol Sasane, *A Friendly Approach to Functional Analysis*, World Scientific Publishing Europe Ltd, 2017.

[3] Robert E. Megginson, *An Introduction to Banach Space Theory*, Springer-Verlag New York Inc, 1998.

[4] Theodore W. Gamelin, *Complex Analysis*, Undergraduate Texts in Mathematics, 2001.

[5] Walter Rudin, *Principles of Mathematical Analysis*, MCGraw-Hill, 1976.

[6] Stephen H. Friedberg, *Linear Algebra*, Pearson Education, Inc, 2019.

[7] Kristopher Tapp, *Matrix Groups for Undergraduates*, American Mathematical Society, 2005.

[8] Leonard Euler, General formulas for any translation of rigid bodies, https://www.17centurymaths.com/contents/euler/e478tr.pdf, 1775. Accessed: 2025-03-11

[9] David Cherney, *Linear Algebra*, UCDavis, 2013.